

Medicine

at Swansea University

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Introduction

This document contains my notes from Swansea University's graduate-entry medicine course. The notes are accessible through GitHub:

<https://github.com/jfbeda/medicine>

You are encouraged to edit and contribute to the notes as you see fit. My contact details are available on my website:

<https://beda.ca/jack>

0.1 Organisation

The main body of the handbook is divided into chapters. Each chapter is stored as a separate file inside the **chapters** folder.

For example:

- `chapters/02-cardiovascular.tex`
- `chapters/03-respiratory.tex`
- `chapters/04-gastrointestinal.tex`

The chapters are imported in the required order from `main.tex`.

CHAPTER 1

Administration

This chapter contains all of the initial administration processes I needed to do, and the order in which I did them.

1.1 Learning@Wales Training

1.1.1 Registering for Learning@Wales

1. Go to learning.nhs.wales.
2. Select **Learning@Wales**.
3. Select the button right below the text that says

‘Please click the button below to register for an account if you are unable to self-register.’

4. Register, and wait to be accepted.

1.1.2 Required training in Learning@Wales

Complete the following training modules:

- [Information Governance, Records Management & Cyber Security](#)
- Fire Safety
- [Infection Prevention & Control Level 1](#)
- Health & Safety
- Treat Me Fairly
- Safeguarding People

- Violence Against Women
- Violence and Aggression
- Manual Handling
- ANTT

1.2 Canvas training

- [NAC - Anti-racism Course](#)
- [NAC - Enhance your EDI Skills](#)

1.3 EOLAS App

- Download the EOLAS app.
- Choose ‘SBUHB Undergraduate Medical Education’ as your organisation.
- Choose ‘SUHB Year 1 Students’ for ‘Is there a team you work most closely with?’

CHAPTER 2

Resources and Useful Links

2.1 Administration links

- [Swansea GEM portal](#)
 - [Student feedback](#)
 - [Leave of absence request](#)
- [International guidance on paying tuition fees](#)
- [Intranet Student profile](#)

2.2 Resource booking

- [Library group study room booking](#)
- [Clinical skills room \(Grove 227\) booking](#)

2.3 Learning resources

- [Complete anatomy](#) (activation code: 83548XFMGSQJ)
- [Dissectable Me](#) (5 minute anatomy podcasts)
- [Kenhub](#) (anatomy learning platform)
- [Teach Me Anatomy](#) (anatomy learning platform)
- [Swansea GEM anatomy spotter quiz](#)
- [Arterial blood gas interpretation website](#)

CHAPTER 3

Useful Contacts

- **Kenny McKeegan**
- **Carl Rowe**
 - Academic Head of GEM / Programme Director
 - **Office:** SUSIM building?
- **Christian Cobbold**
 - **Office:** Grove building, room 243, near the GEM common room.
- **Brendan Mason**
 - Head of literature review
 - Happy to be emailed again if you haven't received a reply after 48 h
- **Karpagam Krishnamoorthy**
 - **Office:** Grove 154A
- **Lauren Blake**
 - Head of RRHiME
- **Llinos Roberts**
 - part of CBL team

CHAPTER 4

Anatomy

4.1 Anatomical terms

4.1.1 Anatomical planes

- Coronal (frontal) plane: divides the body into anterior and posterior halves.
- Sagittal plane: divides the body into left and right halves. The **median plane** divides the body into equal left and right halves, while a **parasagittal plane** divides the body into unequal left and right halves.
- Transverse (horizontal or axial) plane: divides the body into superior and inferior halves.

4.1.2 Anatomical positional terms

- Superior
- Inferior
- Cephalic (towards the head)¹
- Caudal (towards the tail)¹
- Proximal (closer to the point of attachment)
- Distal (further from the point of attachment)
- Medial
- Lateral
- Posterior/dorsal
- Anterior/ventral

¹The terms **cephalic** and **caudal** are often used in embryology, where the head and tail of the embryo are more distinct than in adults.

4.1.3 Anatomical movement terms

- Supination: rotation of the forearm so that the palm faces anteriorly (or superiorly when the arm is outstretched)
- Pronation: rotation of the forearm so that the palm faces posteriorly (or inferiorly when the arm is outstretched)
- Inversion: rotation of the foot so that the sole faces medially
- Eversion: rotation of the foot so that the sole faces laterally
- Flexion: movement that decreases the angle between two body parts²
- Extension: movement that increases the angle between two body parts²
- Abduction: movement away from the midline of the body
- Adduction: movement towards the midline of the body
- Circumduction: circular movement of a body part, combining flexion, extension, abduction, and adduction
- Rotation: movement around a longitudinal axis, either medially (internal rotation) or laterally (external rotation)
- Elevation: movement of a body part superiorly
- Depression: movement of a body part inferiorly
- Protraction: movement of a body part anteriorly
- Retraction: movement of a body part posteriorly
- Opposition: movement of the thumb to touch the tips of the other fingers
- Reposition: movement of the thumb back to its anatomical position
- Dorsiflexion: movement of the foot so that the toes point superiorly
- Plantar flexion: movement of the foot so that the toes point inferiorly
- Palmar flexion: movement of the hand so that the palm faces inferiorly

² the 'angle' being referred to here is the angle in the 'natural' movement (e.g. at the hip it is the raising of the leg that decreases the angle on the anterior side, which is the side that is usually considered when describing flexion and extension).

4.1.4 Anatomical resources

- [Scrollable axial CT](#)

4.2 Stomach

4.2.1 Cells of the stomach

Cell	Purpose	Location
Gastric parietal cells	Releases gastric acid and intrinsic factor	Fundus and body of the stomach
G-cells	Release gastrin	Primarily in the antrum
D-cells	Produce somatostatin	Stomach/intestine/pancreatic islets
ECL-like cells	Produce histamine	Entire stomach
EC cells	Produce serotonin	Whole digestive tract
P/D1 cells	Produce ghrelin	The fundus
Chief cells	Releases pepsinogen and gastric lipase	The stomach

Table 4.1: Gastrointestinal cell types.

4.2.2 Hormones of the Stomach

Substance	Effect	Released by	Release induced by
Gastrin	↑ HCl release by stimulating parietal cells and stimulating ECL-like cells to release histamine.	G cells	—
Histamine	↑ HCl release by binding to receptors on parietal cells.	ECL-like cells	—
Somatostatin	↓ HCl release and ↓ gastrin release.	D cells	Low pH (high acidity)
Serotonin	Regulates GI motility and fluid secretion.	EC cells	—
Ghrelin	Increases appetite and promotes fat storage.	P/D1 cells	—

Table 4.2: Gastrointestinal signalling substances and their effects.

Note that CCK (cholecystokinin) is released by I cells in the duodenum and jejunum.

CHAPTER 5

The Immune System

5.1 Complement system

This section is based primarily on [1] and the week 105 lectures from Kenny McKeegan.

The complement system is a part of the innate immune system based on proteins. The proteins that play a role in the complement system are named C1, C2, ...C9 after the order in which they were discovered. Each of the complement system proteins comprises two halves: an **anaphylatoxin** part (labelled with an 'a'), which recruits neutrophils and other phagocytes to the site of infection, and a binding part, labelled with a 'b' that binds to the surface of a pathogen.

The general idea of the complement system is that one of the binding proteins (e.g. C1b) will bind to the surface of a pathogen, which will cause the binding and cleavage of further proteins, causing the binding and cleavage of further proteins, resulting in a **complement cascade**. There are three named kinds of complement cascades covered in the following subsections, but they have many similarities. In particular, all three cascades — at some point in their execution — produce a C3b (often via a **C3 convertase**: a protein that cleaves C3 into C3a and C3b). Once C3b is produced, it binds to the pathogen and **opsonizes** it, among other things, it and the cascades become the same.

In particular, once C3b has bonded to the pathogen, the cascade continues as follows:

- **Factor B** in the plasma binds to C3b, which is converted by **Factor D** into C3bBb on the pathogen surface.
- This C3bBb complex is stabilized by **properdin** and acts as a C3 convertase causing a positive feedback loop.
- Some of the C3bBb is bonded by another C3b to form C3b₂Bb: a C5 convertase.

- The C3b₂Bb cleaves C5 into C5a and C5b. The C5b binds to the protein, attracting C6, C7, and C8 to bind on top.
- The C8 that binds on top allows for up to 16 individual C9 proteins to insert into the pathogen membrane forming a **membrane attack complex (MAC)**.

5.1.1 Classical pathway

The classical pathway begins C1 binding to a pathogen that has already been tagged with appropriate antibodies. The C1 protein of three other proteins called C1q, C1r, and C1s. It is the C1s that cleaves C4 into C4a and C4b and the C4b that results binds to the pathogen. After, the C1s now cleaves C2 into C2a and C2b and the C2b binds onto the C4b forming C4b2b. This C4b2b is a C3 convertase.

5.1.2 Alternative pathway

Compared with the classical pathway, the alternative pathway does not require antibodies. It begins with the spontaneous cleavage of C3 into C3a and C3b.

5.1.3 Lectin pathway

The lectin pathway begins when **mannose-binding lectin (MBL)** or ficolins bind to characteristic carbohydrates on the surface of a pathogen. This activates **MBL-associated serine proteases (MASPs)**, principally MASP-2, which cleave C4 and C2. C4b binds to the pathogen surface and combines with C2b to form C4b2b: a C3 convertase.

5.2 Cytokines and Chemokines

5.3 Antibiotics

5.3.1 What is the physiological use of inflammation?

- Swelling: vasodilated capillaries bring blood to the site of infection.

Antibiotic	Notes
β-Lactam antibiotics (inhibit cell-wall synthesis)	
Phenoxymethylpenicillin	Penicillin used for infections such as streptococcal tonsillitis.
Amoxicillin	Commonly used penicillin, particularly for respiratory infections.
Flucloxacillin	Penicillin used particularly against susceptible <i>Staphylococcus aureus</i> .
Co-amoxiclav	Amoxicillin combined with the β -lactamase inhibitor clavulanic acid.
Cefalexin	A cephalosporin used for selected respiratory, skin and urinary infections.
Macrolides (inhibit protein synthesis)	
Clarithromycin	Used for respiratory infections and sometimes in penicillin allergy.
Erythromycin	Similar to clarithromycin; also used for selected respiratory and skin infections.
Tetracyclines (inhibit protein synthesis)	
Doxycycline	Used for respiratory infections, acne, chlamydia and some atypical infections.
Aminoglycosides (inhibit protein synthesis)	
Gentamicin	Used mainly for serious Gram-negative infections; may cause renal and auditory toxicity.
Fluoroquinolones (inhibit DNA replication)	
Ciprofloxacin	Used for selected Gram-negative infections; use is restricted because of potentially serious adverse effects.
Nitroimidazoles (damage bacterial DNA)	
Metronidazole	Particularly active against obligate anaerobic bacteria.
Antifolates (inhibit folate metabolism)	
Trimethoprim	Used for selected urinary tract infections.
Nitrofurans	
Nitrofurantoin	Used mainly for lower urinary tract infections.
Glycopeptides (inhibit cell-wall synthesis)	
Vancomycin	Used for serious Gram-positive infections, including some MRSA infections.

Table 5.1: Principal classes of antibiotics and selected examples

Cell type	Phagocyte?	Notes
Innate immune cells		
Neutrophils	Yes	Dead neutrophils are a major component of pus. Can differentiate into macrophages or monocyte-derived dendritic cells.
Monocytes	Yes	
Macrophages	Yes	
Dendritic cells	Yes	
Eosinophils	No	
Basophils	No	Granulocytes; releases histamine. Recognise stressed, infected, or malignant cells by a lack of MHC, which triggers killing by apoptosis.
Mast cells	No	
Natural killer (NK) cells	No	
Adaptive immune cells		
B lymphocytes	No	
CD4 ⁺ helper T cells	No	
CD8 ⁺ cytotoxic T cells	No	
Regulatory T cells	No	

Table 5.2: Major cells of the innate and adaptive immune systems. Phagocytes are leukocytes specific to the immune system [CrashCourseImmune2].

- Increased temperature: increases the cell metabolic rate to be able to reproduce faster. [**CrashCourseImmune1**]

5.4 Bacteria

5.4.1 List of bacteria

5.4.2 Oxygen requirements

Obligate anaerobes. Microaerobes are unique in that too much oxygen are toxic, but they require some oxygen. They won't grow in atmospheric oxygen concentrations, but they also don't grow in zero oxygen environments.

The less specific ones are the facultative anaerobes (usually aerobic, but *can* be anaerobic) and facultative aerobes (vice versa)

Anaerobes are particularly important, because after an initial reaction, abscesses will become a low oxygen environment.

Organism/group	Gram	Shape	Example problems
<i>Staphylococcus</i>			
<i>S. aureus</i>	+	Cocci in clusters	Skin infections, pneumonia, sepsis.
<i>S. epidermis</i>	?		Maybe less important
<i>Clostridioides</i>			
<i>C. difficile</i> (C. diff)	+	Spore-forming rods	Antibiotic-associated diarrhoea and colitis.
<i>C. tetani</i>	?		Causes tetanus
<i>C. perfringens</i>	?		
<i>Pseudomonas</i>			
<i>P. aeruginosa</i>	−	Rods	Pneumonia, wound infections and UTIs. Found in the environment, particularly moist areas.
<i>Enterococcus</i>			
<i>E. faecalis</i>	+	Cocci in pairs or chains	UTIs, wound infections and endocarditis.
<i>E. faecium</i>	+	Cocci in pairs or chains	Healthcare-associated infections and bacteraemia.
<i>Streptococcus</i>			
<i>S. pneumoniae</i>	+	Lancet-shaped diplococci	Pneumonia, otitis media and meningitis.
<i>S. mutans</i>	+	Cocci in chains	Normal flora of pharynx and mouth.
<i>Escherichia</i>			
<i>E. coli</i>	−	Rods	UTIs, diarrhoea and sepsis.
<i>Bacteroides</i>			
<i>B. fragilis</i>	−	Rods	
<i>Candida</i>			
<i>C. albicans</i>	NA	Yeast	

Table 5.3: Selected clinically important bacteria arranged by genus. Bold italic headings indicate genera, while indented entries indicate species. A ‘+’ indicates Gram-positive bacteria and a ‘−’ Gram-negative bacteria.

CHAPTER 6

Diarrhea

6.1 The Small Intestine

Based on the video *The small intestine: How does the small intestine absorb so much?* by Christian Cobbold.

- Small intestine has many villi and microvilli to increase surface area for absorption.
- Each epithelial cell contains order 3000 microvilli.
- Glucose is absorbed into cells via sodium-glucose transporters (SGLT1) that work by pumping sodium ions out of the cell (towards the interstitial fluid), creating a sodium gradient (with respect to the intestinal fluid), that results in sodium uptake from the intestinal fluid where glucose is co-transported with sodium into the cell.
- Fat globules are emulsified by bile salts which are then (somehow?) turned into micelles (whatever they happen to be?) which are then turned into water-soluble chylomicrons (diameter order 1 μm) that are absorbed into the lymphatic system.

6.2 The Large Intestine

Based on the video *Large intestine: Absorption and defaecation* by Christian Cobbold.

- Age results in decreased gastrointestinal motility, which can result in constipation.
- Age results in decreased large intestine muscle tone and thus straining when defecating, which can result in haemorrhoids.
- Age results in weakening sphincter muscles which results in acid reflux and heartburn.

6.3 Fluid Movement Between Body Compartments

Based on the video *Fluid movement between body compartments: Why doesn't all your fluid stay in the bloodstream?* by Christian Cobbold.

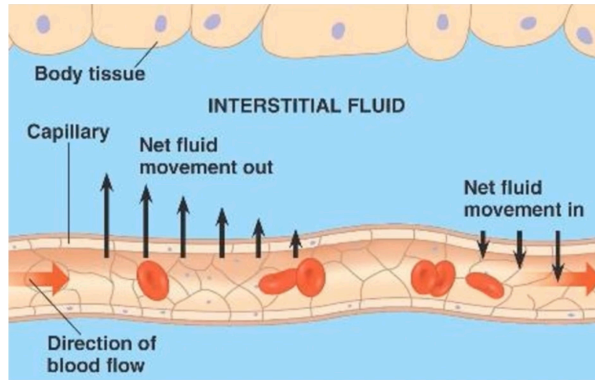


Figure 6.1: Fluid movement between body compartments. [Christian Cobbold]

6.4 Oedema

Based on the video *Oedema: Why fluid accumulates in tissues* by Christian Cobbold.

6.5 2026-09-07 Structure and Function of the GI Tract

By Dr. Christian Cobbold.

CHAPTER 7

Conditions

7.1 Clostridium difficile infection (c. diff)

7.2 Peptic ulcer

Begins with a break in the epithelium which can result from e.g. h. pylori, NSAIDs

30% of ulcers are gastric, and 70% are duodenal [2, 7:32].

7.3 Helicobacter pylori (h. pylori)

Often acquired through oral-oral or oral-fecal in children [2]

h. pylori can activate G cells, which cause the release of more gastrin, and then more gastric acid, which results in mucosal injury [2, t. 5:50].

Gastrin also causes a release of pepsinogen, which turns to pepsin under the influence of acid, which then digests exposed mucosal proteins [2, t. 6:47]. Takeaway message is that pepsin contributes to ulcer deepening, not the cause.

7.4 Gastro-oesophageal reflux disease (GORD)

GORD is any continuous or regular reflux. Anything that disrupts the lower-oesophageal sphincter can lead to GORD

Long term GORD can cause reflux oesophagitis (mucosal redness and itching). Eventually you can get bleeding and scarring in the oesophagus, and then extremely you can get oesophageal narrowing due to scarring and finally Barrett's oesophagus. [3, 9:–]

Causes

- Alcohol and caffeine can cause lower-oesophageal relaxation

CHAPTER 8

Drugs

8.1 Antibiotics

8.1.1 Penicillin

8.1.2 Amoxicillin

8.2 Proton Pump Inhibitors (PPIs)

Examples include

- Omeprazole
- Lansoprazole
- Esomeprazole
- Pantoprazole

PPIs travel through the stomach, and are absorbed in the small intestine into the blood. They act on gastric parietal cells to inhibit the release of gastric acid.

Side effects include:

- Increased risk of *c. diff* infection

8.3 Alginates

Examples include

- Gaviscon

- Peptac
- Wind-eze

Forms a 'raft' over the stomach contents.

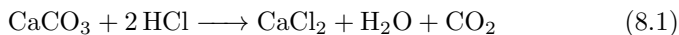
8.4 Antacids

Mechanism of action: neutralizes stomach acid with a chemical reaction.

Examples include:

- Calcium carbonate (CaCO_3)
- Magnesium carbonate (MgCO_3)

For calcium carbonate, the reaction is as follows:



8.5 Prokinetics

Examples include:

- Metoclopramide
- Domperidone
- Erythromycin (off-license)

8.6 H₂-receptor antagonists (H₂RAs)

Mechanism of action: binds to H₂ receptors on gastric parietal cells, preventing histamine from binding and thus reducing gastric acid secretion.

Examples include:

- Famotidine
- Cimetidine

8.7 Non-Steroidal Anti-Inflammatory Drugs (NSAIDs)

Examples include:

- Ibuprofen
- Aspirin
- Naproxen
- Diclofenac

Mechanism of action

After injury, arachidonic acid is released from the cells, which is converted to prostaglandins by COX (Cyclooxygenase) which lead to pain. [2, 10:–]

Prostaglandins produce pain/inflammation/fever. If we inhibit COX-2, we can avoid producing prostaglandins, but as a side effect that prevents the protection of the gastric mucosa.

Dangers

NSAID use increases the risk of peptic ulcers as follows [2, 11:–]. In the stomach, prostaglandins are continuously produced from arachidonic acid, and stimulate the production of mucus and bicarbonate, and help with epithelial cell repair. NSAID consumption then inhibits prostaglandin production and has some knock-on effects.

CHAPTER 9

Acronyms

- NGT: Nasogastric Tube (2026-09-11 lecture with Pramodh V)
- ORS: Oral Rehydration Solution (2026-09-11 lecture with Pramodh V)
- LOCS: Learning Opportunities in Clinical Settings
- CC: Cubic Centimeter
- SAM: Severe Acute Malnutrition (2026-09-11 lecture with Pramodh V)
- MUAC: Mid Upper Arm Circumference (2026-09-11 lecture with Pramodh V) (9.8 cm)
- SCM: Sternocleidomastoid
- PC: Presenting Complaint
- HPC: History of presenting complaint
- GORD: Gastro-oesophageal reflux disease
- GERD: Gastro-esophageal reflux disease
- CNS: Central Nervous System
- PNS: Peripheral Nervous System
- PUD: Peptic Ulcer Disease
- MAC: Membrane Attack Complex
- MALT: Mannose-Binding Lectin *** *****
- MBL: Mannose-Binding Lectin
- PC: Presenting Complaint
- HPC: History of presenting complaint
- NSAID

- PPI
- SSRI
- OAC: Omeprazole, Amoxicillin, Clarithromycin (triple therapy in treatment of *H. pylori*)
- OTC: Over the Counter
- SBA: Single best answer
- PND: Paroxysmal Nocturnal Dyspnea
- SOB: Shortness of Breath
- COPD: Chronic Obstructive Pulmonary Disease
- CP (angle): Costophrenic angle
- CPR: Cardiopulmonary resuscitation
- CXR: Chest X-ray
- ECG: Electrocardiogram
- ABG: Arterial Blood Gas
- LDH: Lactate Dehydrogenase?
- CRP: C-reactive protein
- MT: Medical Thoracoscopy
- VATS: Video-assisted thoracoscopic surgery

CHAPTER 10

Clinical Skills Checklists

10.1 Taking a History

- Presenting Complaint
- History of Presenting Complaint
- Past Medical History
- Drug History
 - Creams, ointments, inhalers, injections, supplements, vitamins
- Allergies
- Family History
 - Genetic conditions, cancers, cardiovascular disease, diabetes, autoimmune conditions
- Social History
 - Occupation
 - Pets (birds)
 - Travel (farmwork)
 - Smoking, alcohol, recreational drugs
- Systems Review

10.2 Abdominal Examination

- Introduction (Name, date of birth, chaperone, any pain)
- End of the bed
 - Bedside equipment, jaundice, pallor, cachexia, distension
- Hands and arms
 - Hands (erythema, Dupuytren's contracture, asterixis, nicotine)
 - Nails (clubbing, koilonychia, leukonychia)
 - Radial pulse
- Face, mouth, neck, and upper chest
 - Eyes (jaundice in sclera, pallor in eyelids)
 - Mouth (angular stomatitis, glossitis, candidiasis, dentation, dehydration)
 - Lymph nodes (cervical, supraclavicular, Virchow's node)
- Chest and Abdomen
 - Spider nevae, striae, scars
 - Cough (hernia)
- Palpation
 - Confirm any pain
 - Nine regions (light then firm palpation)
 - Liver (feel liver edge by moving the hand towards costal margin)
 - Spleen (feel diagonally towards left costal margin)
 - Kidneys (bounce them back and forth)
 - Abdominal aorta
- Percussion
 - Nine regions
 - Liver (upper and lower borders)
 - Ascites (shifting dullness)
- Auscultation
 - Bowel sounds
 - Aortic bruits, renal bruits, iliac/femoral bruits

10.2.1 Respiratory Examination

- Introduction (Name, date of birth, chaperone, any pain)
- End of the bed
 - Bedside equipment, jaundice, pallor, cyanosis,
- Hands and arms
 - Hands (cyanosis, nicotine, asterixis, tremor, temperature)
 - Nails (clubbing, koilonychia)
 - Radial pulse, respiratory rate, blood pressure
- Face, mouth, neck, and upper chest
 - Eyes (pallor, Horner's syndrome)
 - Jugular venous pressure (JVP)
 - Mouth (cyanosis, candidiasis)
 - Trachial deviation
- Lymph nodes
 - Submental, submandibular, pre/post-auricular, superficial/deep cervical, posterior cervical, supraclavicular, oxipital
- Chest and Abdomen
 - Scars, asymmetry, chest shape, use of accessory muscles, respiratory rate, barrel chest
 - Chest expansion (front & back)
 - Chest percussion (front & back)
 - Chest auscultation (front & back)
 - Chest 99 (front & back)
- Oedema
 - Sacral, pedal

10.3 History Taking

Pippa (Doctor) and Katie (Welsh accent)

What can I do for you today? I have some ideas about what we might want to do, but is there anything in particular you were hoping I could do for you? Do you have any idea what caused this? Is there anything you are concerned about? What are you hoping to get out of the appointment today? Sometimes people look things up on the internet, do you have any ideas about what this might be? Was there anything in particular you were hoping to get out of today Smoking? Cannabis? Cocaine? MDMA? Any of the fun stuff?

Presenting complaint SOCRATES Site Onset Character Radiation Associated symptoms Timing Exacerbating and relieving factors Severity

Two children and husband, all well

A week coughing

Green things, first thing in the morning No pain on cough in

DO you have any other medical conditions

CHAPTER 11

Case Histories

This chapter contains fictional clinical scenarios for practising patient-centred history taking. The actor should read the complete brief before the interview but should reveal information only when it is asked for naturally.

11.1 Case 1: Oesophageal Cancer

Name	David Morgan
Age	74 years
Occupation	Retired bus driver
Presenting concern	Increasing difficulty swallowing
Setting	General practice appointment
Intended condition	Oesophageal cancer

Patient identity

You are David Morgan, a 64-year-old retired bus driver. You live with your partner, Helen, and are normally independent. You have come to see the doctor because swallowing has become progressively more difficult.

You are polite and cooperative but visibly worried. You have delayed making the appointment because you hoped the problem would settle by itself.

History of the presenting complaint

Onset: The problem began approximately four months ago. There was no particular event, illness or injury when it started.

Progression: The difficulty has gradually become worse. It was initially occasional but now occurs during most meals.

Solid food: At first, bread and meat felt as though they were sticking. You now avoid most solid food because it is difficult to swallow.

Soft food: During the last few weeks, softer foods such as mashed potato have sometimes also been difficult to swallow.

Liquids: You can still drink liquids. Occasionally they seem to pass slowly, but they do not make you cough or choke.

Location: Food feels as though it sticks behind the lower part of your breastbone. You have no difficulty starting a swallow.

Pain: Swallowing is uncomfortable when food sticks, but it is not usually painful.

Regurgitation: On three or four occasions, a small amount of undigested food has come back into your mouth after it became stuck. You have not had forceful vomiting.

Weight loss: You have unintentionally lost approximately 6 kg in four months. Your weight has fallen from about 82 kg to 76 kg.

Appetite: Your appetite is still present. You eat less because meals take a long time and you are worried about food becoming stuck.

Fatigue: You have felt more tired than usual during the last month, although you can still manage your usual daily activities.

Heartburn: You have experienced intermittent heartburn for several years, usually after a large or spicy meal. You take an over-the-counter antacid when necessary. The heartburn itself has not recently become worse.

Associated symptoms

Reveal the following information only if the student asks:

- No coughing or choking when beginning to swallow.
- No pain in the mouth or throat.
- No persistent nausea or vomiting.
- No blood in vomit.

- No black, tar-like stools and no visible blood in the stool.
- No severe chest pain.
- No persistent abdominal pain.
- No change in voice.
- No persistent cough or breathlessness.
- No weakness, facial drooping or other neurological symptoms.
- No fevers or night sweats.
- No noticeable lumps in the neck.

Past medical and surgical history

Medical history: You have experienced intermittent acid reflux for several years. You were diagnosed with high blood pressure three years ago.

Previous investigation: You have never had a camera test of your oesophagus or stomach.

Surgical history: You had your appendix removed in your twenties. You have had no other operations.

Previous cancer: You have never previously been diagnosed with cancer.

Medication and allergies

Regular medication: Ramipril 5 mg once daily for high blood pressure.

Other medication: An over-the-counter antacid once or twice a week when required. Cocaine user.

Allergies: No known drug allergies.

Family history

Your father died following a heart attack at the age of 71. Your mother died at the age of 83 following a stroke.

You have one older sister, who is generally well. There is no known family history of oesophageal or stomach cancer.

Social history

Home: You live with your partner, Helen, in a two-storey house. You are independent with personal care, cooking and household activities.

Occupation: You retired from bus driving four years ago.

Smoking: You smoked approximately 15 cigarettes per day from the age of 20 until the age of 59. You stopped smoking five years ago.

Alcohol: You usually drink three or four pints of beer over the course of a weekend, amounting to approximately 6–8 units each week.

Diet: You previously ate a varied diet. You now mainly have soup, yoghurt, porridge and other soft foods. You drink water with meals to help food pass.

Recreational drugs: You have never used recreational drugs.

Effect on daily life

Meals now take much longer than usual, and you often leave food unfinished. You have stopped eating meat and bread and have begun avoiding meals in restaurants because you are embarrassed that food might become stuck.

Helen has noticed the weight loss and encouraged you to make this appointment. You remain independent but have less energy for gardening and walking.

You are worried that you will continue losing weight or eventually become unable to swallow anything.

Ideas, concerns and expectations

Ideas: You initially thought the problem was caused by indigestion. You now wonder whether something is narrowing your ‘food pipe’.

Concerns: Your main fear is cancer, particularly because the problem is getting worse and you have lost weight. You are also worried about how Helen would cope if you became seriously ill.

Expectations: You want to know what could be causing the problem and whether you will need a scan or a camera test. You hope that the cause is treatable.

Questions the actor may ask

If there is time, ask one or two of the following:

- ‘What could be causing this?’
- ‘Do you think it might be cancer?’
- ‘What tests will I need?’
- ‘Is there anything I can do about the weight loss?’
- ‘How quickly will I be seen?’

The actor should accept an explanation that further assessment is needed and that the student cannot confirm a diagnosis from the history alone.

Tutor information

This is a fictional teaching case. The pattern of progressive swallowing difficulty, beginning with solid food and later affecting softer food, suggests a mechanical obstruction of the oesophagus. Unintentional weight loss and increasing symptoms are concerning features that require prompt investigation.

The intended diagnosis is oesophageal cancer. It should not be treated as confirmed during the interview because no investigation results have been provided.

Features the student should identify

- Progressive difficulty swallowing over four months.
- Initial difficulty with solid food, followed by softer food.
- Sensation of food sticking behind the breastbone.
- Unintentional loss of 6 kg.
- Reduced food intake despite a preserved appetite.
- Increasing fatigue.
- Longstanding reflux symptoms.
- Previous smoking history.
- Significant anxiety about cancer and the effect of illness on his partner.

Expected clinical approach

The student should recognise that progressive dysphagia with unintentional weight loss requires urgent clinical assessment. A good consultation should include:

- clarification of whether the difficulty is with initiating a swallow or with food passing through the oesophagus;
- assessment of progression, regurgitation, pain, bleeding and nutritional consequences;

- relevant gastrointestinal, respiratory and neurological symptoms;
- medical, medication, family and social histories;
- assessment of smoking and alcohol exposure;
- exploration of ideas, concerns and expectations;
- acknowledgement of the patient's concern without confirming or excluding cancer prematurely;
- explanation that prompt investigation is required, commonly including upper gastrointestinal endoscopy and biopsy where indicated; and
- consideration of nutritional support and safety-netting while investigation is arranged.

Possible discussion points

1. Which features make this an urgent presentation?
2. What alternative diagnoses could produce progressive dysphagia?
3. What physical examination would be appropriate?
4. What initial investigations and referrals should be considered?
5. How should the possibility of cancer be discussed without implying that the diagnosis is already confirmed?
6. What nutritional and psychosocial support may be helpful?

Teaching note

The precise referral pathway, investigations and management should follow current local and national guidance. No tumour location, histological type or stage is specified in this scenario, and these details should not be assumed.

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